Mathematics III_2024_PYQ

Group-A (Very Short Answer Type Questions)

Answer any ten of the following:

(Each carries 1 mark)

1. Find the number of vertices in a graph with 15 edges if each vertex has degree 2.
2. Determine whether the sequence $\{x_n\}$, where $x_n = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots + \frac{1}{n(n+1)}$ converges or not.
3. Is the function $f(x,y) = \begin{cases} \frac{xy}{xy+x-y}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ continuous at (1,2)?
4. Evaluate: $\int_0^{\pi/2} \int_0^2 r\, dr\, d\theta$
5. Write the general solution of the differential equation $p = \cos(y - xp)$, where $p = \frac{dy}{dx}$.
6. Is the function $f(x,y) = \begin{cases} x^2 + y^2 + xy, & (x,y) \neq (2,3) \\ 10, & (x,y) = (2,3) \end{cases}$ continuous at (0,0)?

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7. Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2+n+1}$ converges or not.
8. Find the degree of the homogeneous function $f(x,y) = \frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}}$.
9. Find the value of $\iint dx\, dy$ over the first quadrant of the circle having centre at the origin and radius 4 cm.
10. Evaluate: $\frac{1}{(D-2)(D-3)} e^{2x}$
11. If a graph has 5 vertices and 7 edges, then write the size of its adjacency matrix.
12. Give an example of a bounded sequence which is not convergent.

Group-B (Short Answer Type Questions)

Answer any three of the following:

(Each carries 5 marks)

2. Let $f(x,y) = \begin{cases} \frac{x^3+y^3}{x-y}, & x \neq y \\ 0, & x = y \end{cases}$. Prove that $f(x,y)$ is not continuous at (0,0) but $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist at (0,0).

3. Using the transformation $x + y = u, y = uv$, show that $\int_0^1 dx \int_0^{1-x} e^{\frac{y}{x+y}} dy = \frac{1}{2}(e-1)$.

4. Solve: $x^2 \frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = x^2 \log\, x$.

5. Prove that the maximum degree of any vertex in a simple graph with n vertices is $n-1$.

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Prove that the maximum number of edges in a connected simple graph with n vertices is $\frac{1}{2}n(n-1)$.

6. Using the transformation $x - y = u, x + y = v$, prove that $\iint_R \cos\frac{x-y}{x+y}\, dx\, dy = \frac{\sin 1}{2}$, where R is the region bounded by $x + y = 1, x = 0, y = 0$.

Group-C (Long Answer Type Questions)

Answer any three of the following:

(Each carries 15 marks, each subpart usually 5 marks)

7. (a) Prove that the minimum number of edges in a connected graph with n vertices is $n-1$.

   (b) Prove that a complete graph with n vertices consists of $\frac{1}{2}n(n-1)$ number of edges.

   (c) Show that a bipartite graph cannot contain a cycle of odd length.

   (d) Prove that there exists no simple graph with five vertices having degrees 4, 4, 4, 2, 2.

   Draw, if possible, a simple graph with five vertices having degrees 2, 3, 3, 3, 3.

8. (a) Show that the sequence $\left\{\sqrt{5}, \sqrt{5+\sqrt{5}}, \sqrt{5+\sqrt{5+\sqrt{5}}}, \ldots\right\}$ tends to a definite finite limit. Also find the limit.

(b) Discuss the convergence of the power series $1+\frac{3}{7}x+\frac{3\cdot 6}{7\cdot 10}x^2+\frac{3\cdot 6\cdot 9}{7\cdot 10\cdot 13}x^3+\cdots, x>0$.

(c) Discuss the convergence of the series $x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$, be any number.

9. (a) Let $f(x,y)=\begin{cases}1, & xy\neq 0\\ 0, & xy=0\end{cases}$. Show that the two repeated limits exist at (0,0) and are equal, but the simultaneous limit $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

(b) Let $f(x,y)=\begin{cases}x+y\sin\frac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$. Verify that the simultaneous limit $\lim_{(x,y)\to(0,0)} f(x,y)$ exists and the repeated limit $\lim_{y\to 0}\lim_{x\to 0} f(x,y)$ exists but the repeated limit $\lim_{x\to 0}\lim_{y\to 0} f(x,y)$ does not exist.

(c) Let $f(x,y)=\begin{cases}y\sin\frac{1}{x}+x\sin\frac{1}{y}, & xy\neq 0\\ 0, & xy=0\end{cases}$. Show that at (0,0) the double limit exists, but the repeated limits do not exist.

10. (a) Verify Stoke's theorem for $\vec{F}=(2x-y)\hat{\imath}-yz^2\hat{\jmath}-y^2z\hat{k}$ where S is the upper half surface of the sphere $x^2+y^2+z^2=1$ and C is its boundary.

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(b) Verify Green's theorem in the plane for $\oint_C[(xy+y^2)dx+x^2dy]$ where C is the closed curve of the region bounded by $y=x$ and $y=x^2$.

(c) Verify the Divergence Theorem for the vector function $\vec{F}=(x^2-yz)\hat{\imath}+(y^2-zx)\hat{\jmath}+(z^2-xy)\hat{k}$, taken over the rectangular parallelepiped $0\leq x\leq a, 0\leq y\leq b, 0\leq z\leq c$.

11. (a) Find div $\vec{F}$ and curl $\vec{F}$ where $\vec{F}=\text{grad}(x^3+y^3+z^3-3xyz)$.

(b) Show that $\vec{F}=(6xy+z^3)\hat{\imath}+(3x^2-z)\hat{\jmath}+(3xz^2-y)\hat{k}$ is irrotational. Find a scalar function φ such that $\vec{F}=\nabla\varphi$.

(c) If $r=|\vec{r}|$ where $\vec{r}=x\hat{\imath}+y\hat{\jmath}+z\hat{k}$, then prove that $\nabla\cdot\left(\frac{\vec{r}}{r}\right)=\frac{2}{r}$.